Climate-Shock Financial Response

Spec A (all shocks) vs Spec B (ENSO-attributed) · Flood & Drought · by company

1. Defining the shocks (as in Gleb's master stress ledger)

Common structure

Each channel is a physical THRESHOLD crossing × an ENSO-alignment gate. threshold_crossed = Spec A (general shock). stress_flag = threshold_crossed AND enso_aligned = Spec B (ENSO-attributed). The ENSO gate needs |ONI| ≥ 0.5, the teleconnection ρ·ONI of the channel-relevant sign, and the realised regional anomaly Z moving that way. A year is tagged for a mine if any month carries the flag — read directly off the ledger.

FLOOD — Spec A / Spec B

Threshold = 20 mm precipitation (Wilcox 2016): the area-weighted regional-mean monthly accumulation ≥ 20 mm/month (FLOOD_regmean_M, the headline). ENSO gate: ρ_pr·ONI > 0 and realised z_pr > 0 — the wetter, El Niño direction.

DROUGHT — Spec A / Spec B

Threshold = mine-cell SPEI-12 ≤ −1 (WMO moderate-drought; DROUGHT_meteo, the headline; missing SPEI = unknown, not zero). ENSO gate: ρ_pr·ONI < 0 and realised z_pr < 0 — the drier direction (sign flips vs flood). Gleb's regressor is the 3-yr cumulative ENSO-aligned share cum3_str_meteo; here we report the yearly tag (any flagged month) for the incidence view.

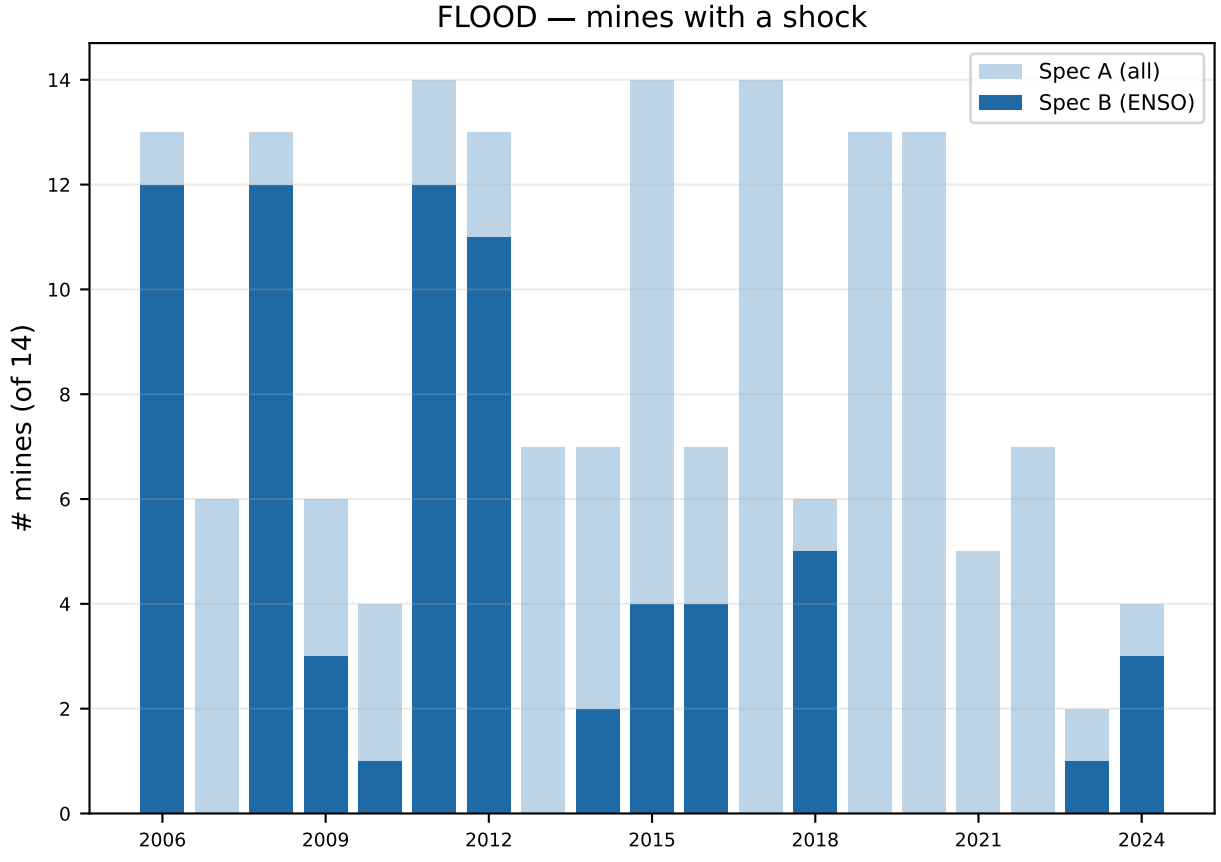
Desalination flag

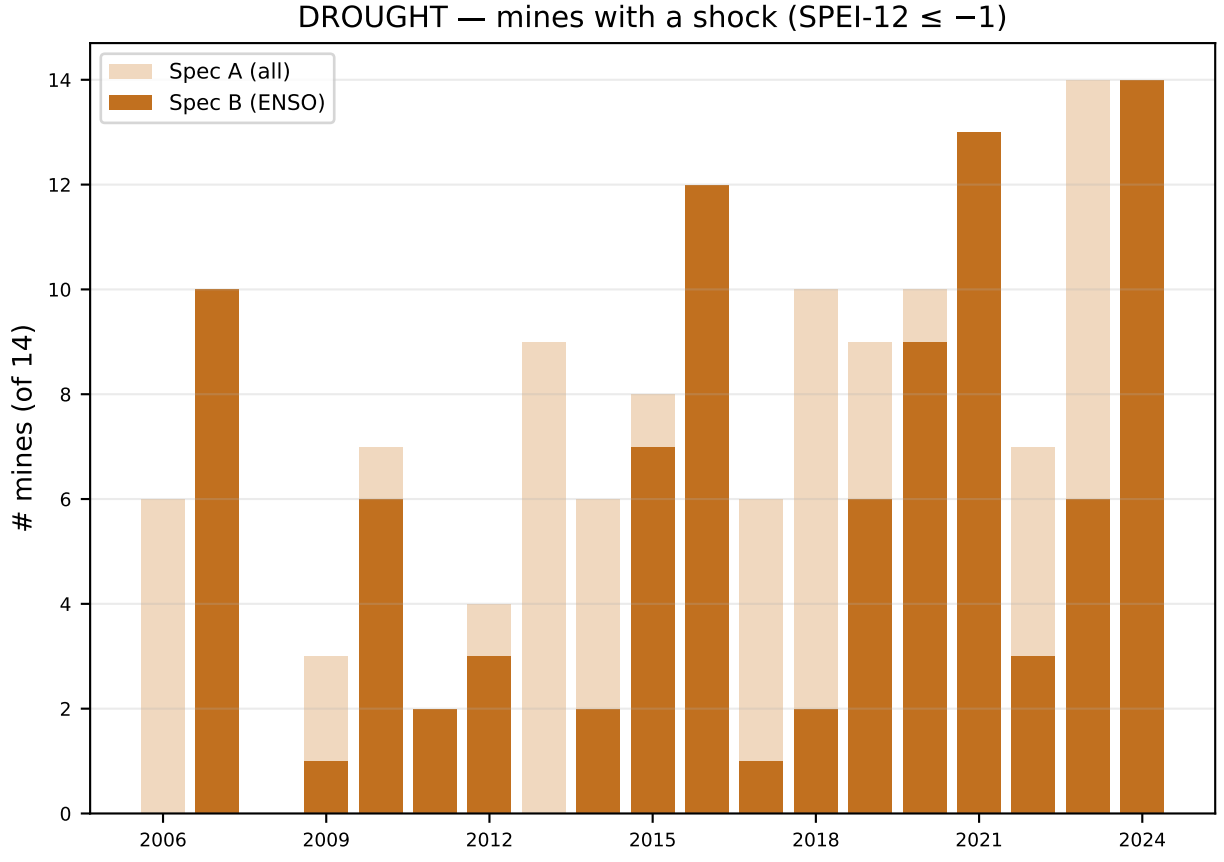
From mines_desal.csv (time-varying). Desalinated from: Centinela 2010, Candelaria 2013, Escondida 2018, Spence 2021, Collahuasi & Quebrada Blanca 2023, Los Pelambres 2024. Codelco, Anglo American Sur, Cerro Colorado, Zaldívar, El Abra, Lomas Bayas, Mantos stay continental.

Financial reading

Canonical statements (USD millions). We read what moved in shock years — revenue, OPEX, net income — and, to test whether operators borrowed their way through, current liabilities, non-current liabilities (long-term debt), cash, and operating cash flow.

How many mines carry a shock each year — Spec A vs Spec B

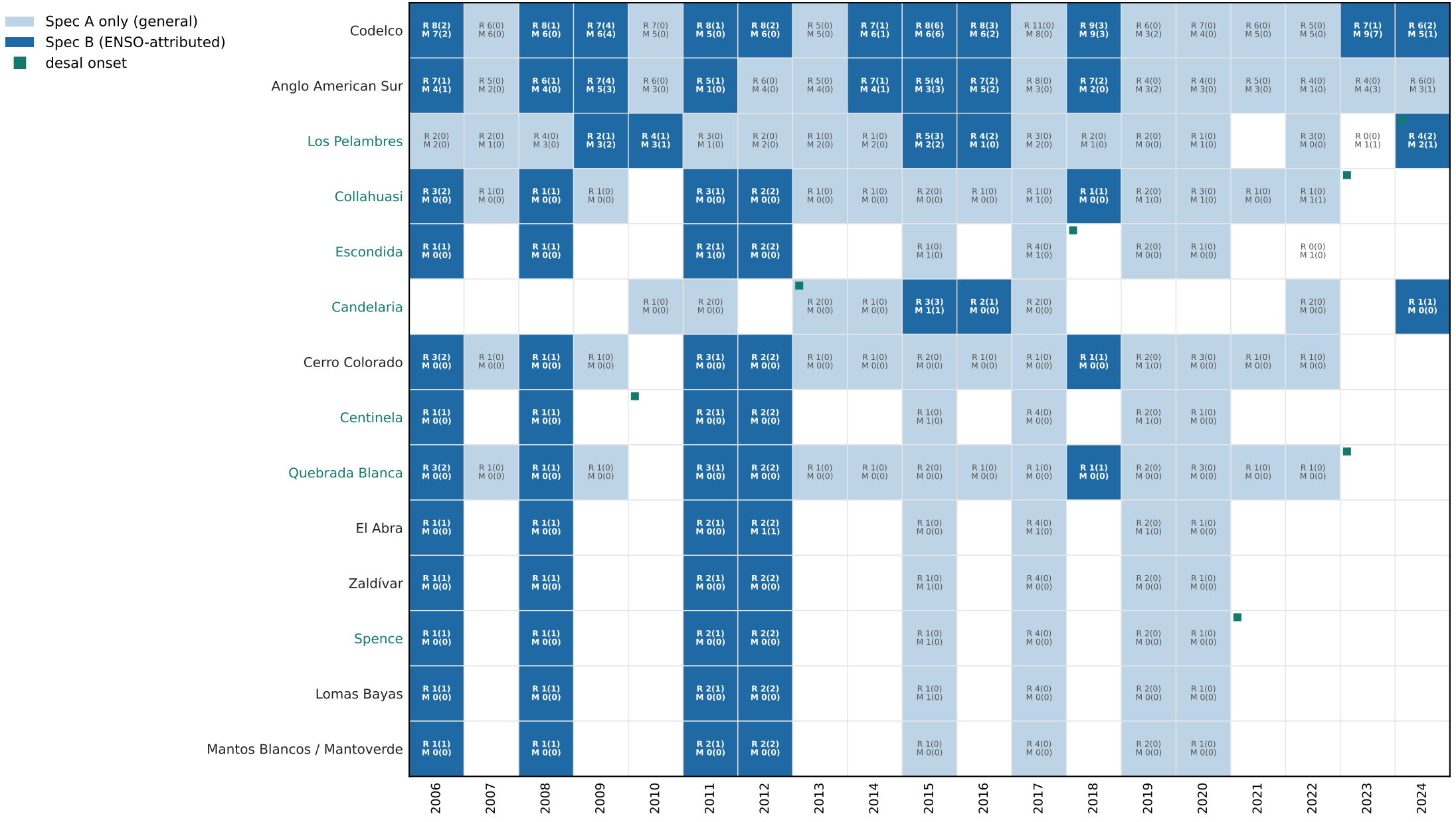




aveat: 60% of ENSO-attributed flood mine-years fire in only a single month — too brief to expect a clean annual-statement signature. The flood grid (next page) shows the per-cell firing count; the financial reading leans on multi-month flood year

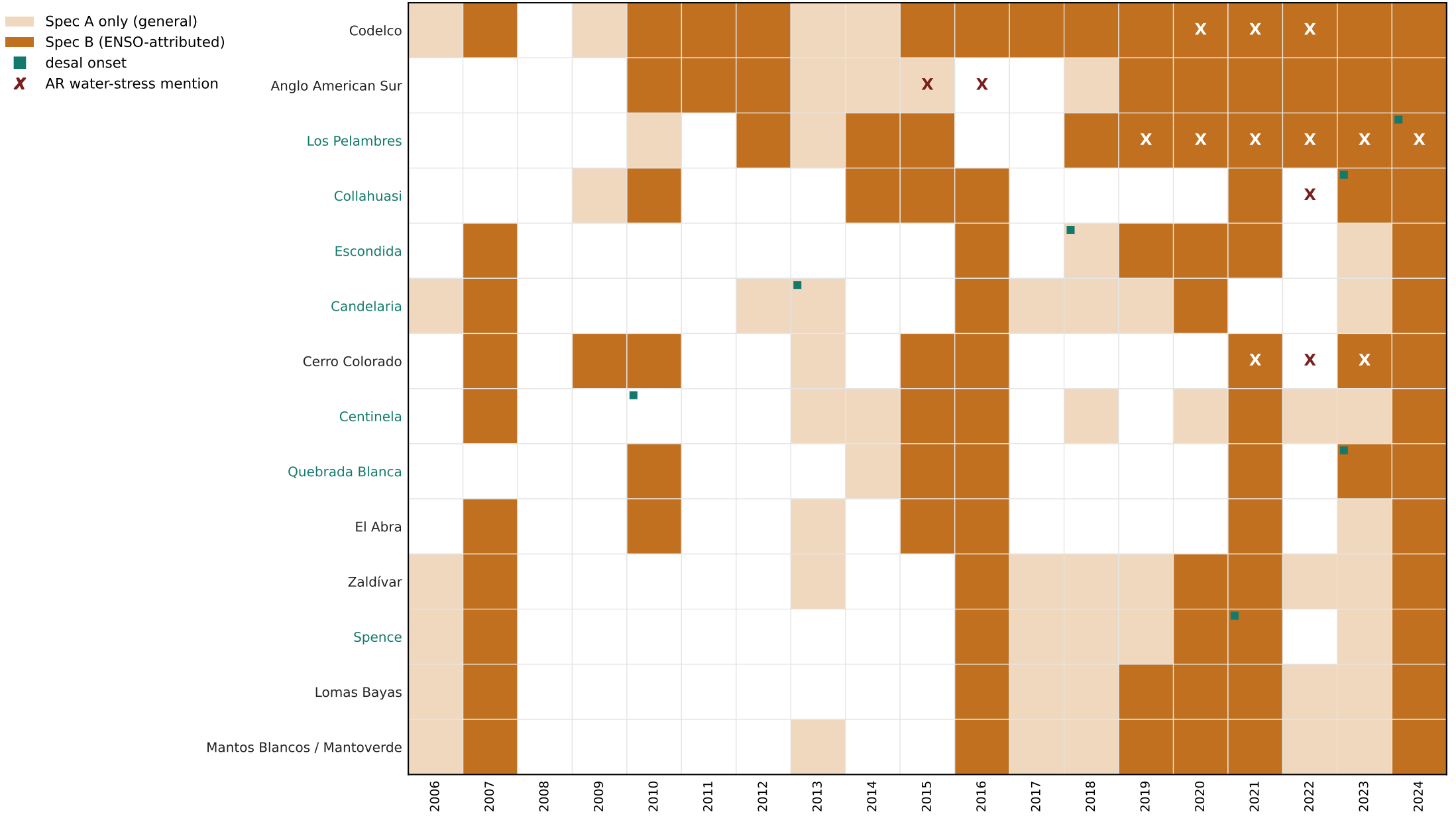
FLOOD shock grid — firing-months, general count with ENSO-attributed in (parentheses)

Cell: R = regional-mean firing-months, M = mine-cell (daily-threshold) firing-months; ENSO-attributed in (parens). Master ledger (monthly). Row label teal = desalinated (■ desal-onset); black = continental



DROUGHT shock grid (Spec A light · Spec B dark)

X = annual-report water-stress mention (Rohan). Row label teal = desalinated (■ desal-onset); black = continental



Climate hits vs annual-report water-stress mentions

By company-year, 2014–2024. Flood hits = ENSO (all) firing-months; Drought = ENSO / general; AR mention = exact wording from Rohan (CHC_FinancialSignals_WaterStress).
Rows shown only where a flood-ENSO, drought-ENSO, or AR signal is present.

Mine	Year	Flood hits ENSO (all) mo	Drought hit	AR water-stress mention — exact wording
Codelco	2014	1 (7)	gen	—
	2015	6 (8)	ENSO	—
	2016	3 (8)	ENSO	—
	2017	0 (11)	ENSO	—
	2018	3 (9)	ENSO	—
	2019	0 (6)	ENSO	—
	2020	0 (7)	ENSO	"La Niña 2020-22 drought: water restrictions on Pampa Tamarugal / Loa River system" [DGA]
	2021	0 (6)	ENSO	(same wording as above)
	2022	0 (5)	ENSO	(same wording as above)
	2023	1 (7)	ENSO	—
	2024	2 (6)	ENSO	—
Anglo American Sur	2014	1 (7)	gen	—
	2015	4 (5)	gen	"Maipo River basin stress; El Niño 2015-16 reduced flows; Santiago downstream scrutiny" [DGA]
	2016	2 (7)	—	(same wording as above)
	2018	2 (7)	gen	—
	2019	0 (4)	ENSO	—
	2020	0 (4)	ENSO	—
	2021	0 (5)	ENSO	—

Green row = AR mention coincides with a climate hit (triangulated). Orange = AR mention with no Spec-A/B climate hit that year.

Climate hits vs annual-report water-stress mentions (cont.)

By company-year, 2014–2024. Flood hits = ENSO (all) firing-months; Drought = ENSO / general; AR mention = exact wording from Rohan (CHC_FinancialSignals_WaterStress).
Rows shown only where a flood-ENSO, drought-ENSO, or AR signal is present.

Mine	Year	Flood hits ENSO (all) mo	Drought hit	AR water-stress mention — exact wording
	2022	0 (4)	ENSO	—
	2023	0 (4)	ENSO	—
	2024	0 (6)	ENSO	—
Los Pelambres	2014	0 (1)	ENSO	—
	2015	3 (5)	ENSO	—
	2016	2 (4)	—	—
	2018	0 (2)	ENSO	—
	2019	0 (2)	ENSO	"Severe Coquimbo drought; Choapa River flows reduced; community blockades 2022; Plan Sequía implemented" [DGA / 1TA]
	2020	0 (1)	ENSO	(same wording as above)
	2021	—	ENSO	(same wording as above)
	2022	0 (3)	ENSO	(same wording as above)
	2023	—	ENSO	(same wording as above)
	2024	2 (4)	ENSO	(same wording as above)
Collahuasi	2014	0 (1)	ENSO	—
	2015	0 (2)	ENSO	—
	2016	0 (1)	ENSO	—
	2018	1 (1)	—	—
	2021	0 (1)	ENSO	—

Green row = AR mention coincides with a climate hit (triangulated). Orange = AR mention with no Spec-A/B climate hit that year.

Climate hits vs annual-report water-stress mentions (cont.)

By company-year, 2014–2024. Flood hits = ENSO (all) firing-months; Drought = ENSO / general; AR mention = exact wording from Rohan (CHC_FinancialSignals_WaterStress).
Rows shown only where a flood-ENSO, drought-ENSO, or AR signal is present.

Mine	Year	Flood hits ENSO (all) mo	Drought hit	AR water-stress mention — exact wording
	2022	0 (1)	—	"Tarapacá region water restrictions; Salar de Coposa wetland monitoring; desal construction started 2022" [DGA]
	2023	—	ENSO	—
	2024	—	ENSO	—
Escondida	2016	—	ENSO	—
	2019	0 (2)	ENSO	—
	2020	0 (1)	ENSO	—
	2021	—	ENSO	—
	2024	—	ENSO	—
Candelaria	2015	3 (3)	—	—
	2016	1 (2)	ENSO	—
	2020	—	ENSO	—
	2024	1 (1)	ENSO	—
Cerro Colorado	2015	0 (2)	ENSO	—
	2016	0 (1)	ENSO	—
	2018	1 (1)	—	—
	2021	0 (1)	ENSO	"Lagunillas aquifer (Pampa Tamarugal) halt ordered Sept 2021 (150 days); 2nd restriction 2022; RCA expired Dec 2023 → permanent closure" [1er Tribunal Ambiental de Antofagasta]
	2022	0 (1)	—	(same wording as above)
	2023	—	ENSO	(same wording as above)

Green row = AR mention coincides with a climate hit (triangulated). Orange = AR mention with no Spec-A/B climate hit that year.

Climate hits vs annual-report water-stress mentions (cont.)

By company-year, 2014–2024. Flood hits = ENSO (all) firing-months; Drought = ENSO / general; AR mention = exact wording from Rohan (CHC_FinancialSignals_WaterStress).
Rows shown only where a flood-ENSO, drought-ENSO, or AR signal is present.

Mine	Year	Flood hits ENSO (all) mo	Drought hit	AR water-stress mention — exact wording
	2024	—	ENSO	—
Centinela	2015	0 (1)	ENSO	—
	2016	—	ENSO	—
	2021	—	ENSO	—
	2024	—	ENSO	—
Quebrada Blanca	2015	0 (2)	ENSO	—
	2016	0 (1)	ENSO	—
	2018	1 (1)	—	—
	2021	0 (1)	ENSO	—
	2023	—	ENSO	—
	2024	—	ENSO	—
El Abra	2015	0 (1)	ENSO	—
	2016	—	ENSO	—
	2021	—	ENSO	—
	2024	—	ENSO	—
Zaldívar	2016	—	ENSO	—
	2020	0 (1)	ENSO	—
	2021	—	ENSO	—

Green row = AR mention coincides with a climate hit (triangulated). Orange = AR mention with no Spec-A/B climate hit that year.

Climate hits vs annual-report water-stress mentions (cont.)

By company-year, 2014–2024. Flood hits = ENSO (all) firing-months; Drought = ENSO / general; AR mention = exact wording from Rohan (CHC_FinancialSignals_WaterStress).
Rows shown only where a flood-ENSO, drought-ENSO, or AR signal is present.

Mine	Year	Flood hits ENSO (all) mo	Drought hit	AR water-stress mention — exact wording
	2024	—	ENSO	—
Spence	2016	—	ENSO	—
	2020	0 (1)	ENSO	—
	2021	—	ENSO	—
	2024	—	ENSO	—
Lomas Bayas	2016	—	ENSO	—
	2019	0 (2)	ENSO	—
	2020	0 (1)	ENSO	—
	2021	—	ENSO	—
	2024	—	ENSO	—
Mantos Blancos / Mantoverde	2016	—	ENSO	—
	2019	0 (2)	ENSO	—
	2020	0 (1)	ENSO	—
	2021	—	ENSO	—
	2024	—	ENSO	—

Green row = AR mention coincides with a climate hit (triangulated). Orange = AR mention with no Spec-A/B climate hit that year.

Drought (ENSO) shocks — every shock year by mine

Year-on-year change in each item for every ENSO-attributable drought year. Desal?: post = on desalinated seawater that year, pre = before its plant, — = continental.

Mine (desal onset)	Year	Desal?	OPEX	CAPEX	Prod	C1	Cur.Liab	Cash	Closure
Codelco	2007	—	+10%	+32%	—	—	+59%	+157%	—
	2010	—	+21%	+37%	—	+12%	+28%	+13%	-6%
	2011	—	+13%	-3%	—	+11%	-16%	+58%	-3%
	2012	—	+3%	+64%	-0%	+40%	-6%	-9%	-38%
	2015	—	-2%	+12%	+5%	-8%	+4%	+33%	—
	2016	—	-5%	-29%	-4%	-9%	-34%	-67%	-97%
	2017	—	+10%	+13%	+9%	+8%	+35%	+151%	-100%
	2018	—	+8%	+14%	-0%	+2%	+7%	-15%	—
	2019	—	-10%	+5%	-7%	+2%	+11%	+6%	—
	2020	—	+5%	-42%	+1%	-9%	-12%	+62%	+0%
	2021	—	+15%	+18%	+1%	+3%	+15%	-39%	-100%
	2022	—	+1%	+23%	-17%	+25%	-0%	-20%	—
	2023	—	+8%	+26%	-8%	+23%	+12%	+31%	—
	2024	—	-3%	+10%	+4%	-2%	+13%	-49%	—
Anglo American Sur	2010	—	+7%	+72%	+2%	—	+39%	+13%	+13%
	2011	—	+32%	-11%	+1%	—	-4%	—	+16%
	2012	—	+45%	-58%	—	—	+104%	+40%	-1%
	2019	—	-7%	+20%	-65%	-7%	-40%	-33%	+36%
	2020	—	+24%	+10%	+162%	+10%	+63%	+62%	+48%
	2021	—	-2%	+83%	-66%	+6%	+30%	+9%	-28%
	2022	—	+19%	+33%	—	+35%	-8%	-50%	+28%
	2023	—	+2%	-25%	—	+42%	-3%	-80%	+17%
	2024	—	-15%	-46%	—	-10%	-12%	+159%	-8%
Los Pelambres (desal 2024)	2012	pre	+7%	+24%	—	+10%	-9%	+96%	+1%
	2014	pre	-13%	+10%	—	+2%	+40%	+76%	-43%
	2015	pre	-16%	-18%	—	+4%	—	+14%	+5%

Drought (ENSO) shocks — every shock year by mine (cont.)

Mine (desal onset)	Year	Desal?	OPEX	CAPEX	Prod	C1	Cur.Liab	Cash	Closure
	2018	pre	+9%	+51%	+4%	-11%	-47%	+95%	-4%
	2019	pre	-4%	+48%	+1%	+0%	+16%	-82%	-21%
	2020	pre	-21%	+54%	-1%	-11%	+98%	+195%	+31%
	2021	pre	+11%	+8%	-10%	+10%	-26%	-94%	-38%
	2022	pre	-7%	+7%	-15%	+24%	+43%	—	+35%
	2023	pre	+14%	-5%	+9%	+4%	+21%	-11%	-2%
	2024	**post**	+38%	-7%	+7%	+11%	+59%	+124%	-8%
Collahuasi (desal 2023)	2010	pre	-2%	+15%	—	—	+52%	+33%	+0%
	2014	pre	+8%	-46%	+2%	—	-32%	+35%	+4%
	2015	pre	-7%	-25%	+4%	—	-32%	+34%	-3%
	2016	pre	-9%	+36%	+5%	-34%	+51%	-24%	+11%
	2021	pre	+5%	+13%	+2%	-2%	+19%	+257%	-49%
	2023	**post**	+10%	+55%	+2%	+30%	+32%	-27%	-20%
	2024	**post**	+3%	+28%	-2%	+6%	+25%	+59%	+22%
Escondida (desal 2018)	2007	pre	—	—	—	—	—	—	—
	2016	pre	-25%	-48%	+8%	+5%	-40%	+127%	+31%
	2019	**post**	+8%	+7%	-4%	+7%	+11%	-18%	+2%
	2020	**post**	-24%	-33%	-0%	-11%	+2%	+92%	+41%
	2021	**post**	+4%	+4%	-17%	-1%	+8%	-21%	-10%
	2024	**post**	+10%	+33%	—	+4%	+47%	+79%	-6%
Candelaria (desal 2013)	2007	pre	—	—	—	—	—	—	—
	2016	**post**	-6%	-36%	—	+5%	+46%	+50%	+33%
	2020	**post**	—	—	-15%	-6%	—	—	—
	2024	**post**	+6%	-27%	—	-16%	+94%	+67%	-5%
Cerro Colorado	2007	—	+18%	+289%	—	—	-21%	+400%	+206%
	2009	—	-1%	-20%	—	—	+2%	+6%	+5%

Drought (ENSO) shocks — every shock year by mine (cont.)

Mine (desal onset)	Year	Desal?	OPEX	CAPEX	Prod	C1	Cur.Liab	Cash	Closure
	2010	—	—	+18%	—	—	+29%	+11%	+24%
	2015	—	-30%	+44%	—	—	-13%	—	-1%
	2016	—	-35%	-18%	-7%	—	+12%	—	+23%
	2021	—	—	-54%	-9%	—	-37%	-9%	+53%
	2023	—	-41%	-75%	-31%	—	-59%	-57%	+20%
	2024	—	-74%	-100%	—	—	+60%	-62%	+1%
Centinela (desal 2010)	2007	pre	—	—	—	—	—	—	—
	2015	**post**	-7%	+5%	-3%	+13%	+57%	-31%	-6%
	2016	**post**	-24%	+15%	+31%	-36%	+1%	+3%	+11%
	2021	**post**	+8%	+79%	+1%	-11%	-35%	-48%	-43%
	2024	**post**	+1%	+29%	—	-2%	+1%	+427%	+11%
Quebrada Blanca (desal 2023)	2010	pre	+39%	+276%	—	—	+142%	+33%	+3%
	2015	pre	-2%	-15%	—	—	+26%	-24%	—
	2016	pre	-11%	-43%	—	—	+10%	-78%	+188%
	2021	pre	—	+67%	+10%	—	+5%	+8%	+42%
	2023	**post**	—	-7%	+161%	—	-18%	+32%	+3%
	2024	**post**	+212%	-57%	+24%	—	-25%	-9%	+88%
El Abra	2007	—	-3%	—	—	—	-67%	—	—
	2010	—	+60%	—	—	—	+81%	-7%	-8%
	2015	—	+9%	-21%	-3%	—	-54%	-12%	-26%
	2016	—	-46%	-92%	-9%	—	-11%	+249%	-6%
	2021	—	+8%	-27%	—	—	+161%	+176%	-13%
	2024	—	-5%	-20%	—	—	+13%	+82%	-19%
Zaldívar	2007	—	—	—	—	—	—	—	—
	2016	—	+203%	+380%	+275%	-10%	-27%	+483%	+44%
	2020	—	-13%	+66%	-17%	+3%	+143%	-52%	+15%

Drought (ENSO) shocks — every shock year by mine (cont.)

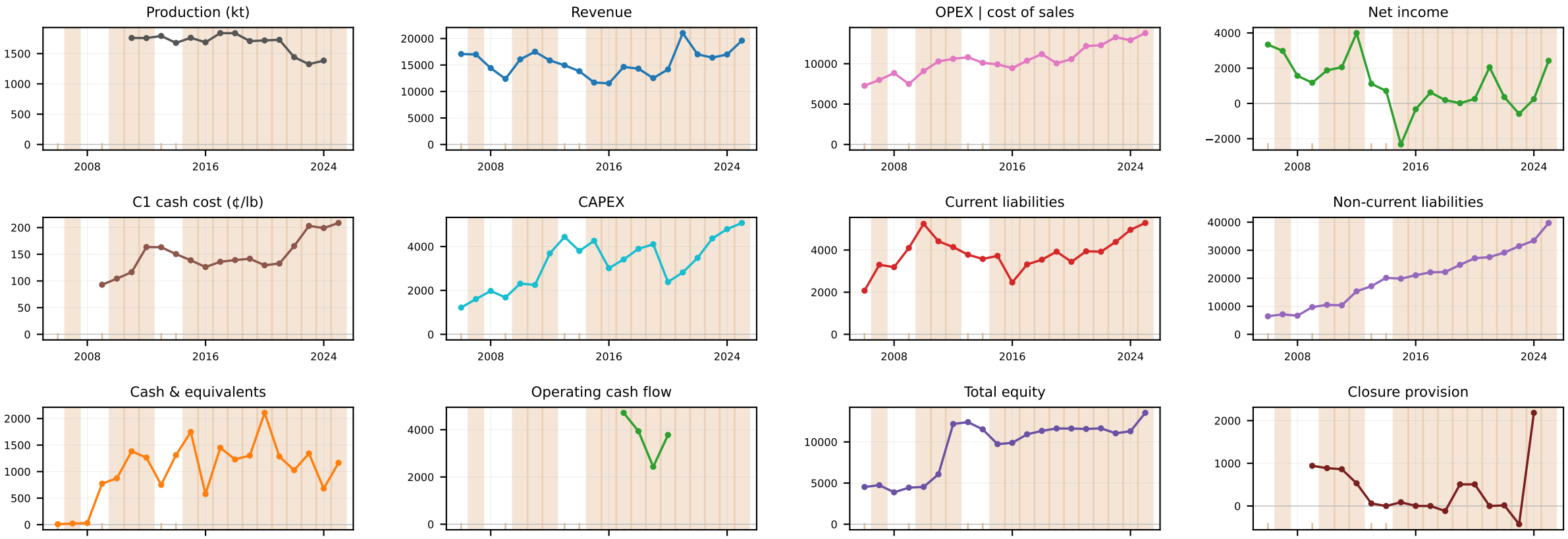
Mine (desal onset)	Year	Desal?	OPEX	CAPEX	Prod	C1	Cur.Liab	Cash	Closure
	2021	—	+23%	+17%	-9%	+33%	-25%	-30%	-31%
	2024	—	-4%	-4%	-1%	+2%	-20%	+152%	-13%
Spence (desal 2021)	2007	pre	—	—	—	—	—	—	—
	2016	pre	+28%	+4%	+4%	—	-13%	—	+56%
	2020	pre	—	-25%	-24%	—	-13%	+41%	+252%
	2021	**post**	—	-26%	+0%	—	+18%	-20%	+22%
	2024	**post**	+10%	-8%	—	+1%	+23%	-54%	-0%
Lomas Bayas	2007	—	—	—	—	—	-45%	-56%	—
	2016	—	-12%	-32%	+0%	—	-27%	-61%	+14%
	2019	—	—	—	—	—	—	—	—
	2020	—	—	—	—	—	—	—	—
	2021	—	—	—	—	—	—	—	—
	2024	—	+15%	+116%	—	—	+24%	-80%	-45%
Anglo Am. Norte / Capstone	2007	—	—	—	—	—	—	—	—
	2016	—	-13%	-25%	—	—	+10%	+0%	-1%
	2019	—	—	—	—	—	—	—	—
	2020	—	—	—	—	—	—	—	—
	2021	—	—	—	—	—	—	—	—
	2024	—	—	—	+12%	—	—	—	—

Current liabilities are short-term funding: at the desalinating mines the rise largely reflects construction-period payables for the build, not drought financing (multi-year capex is raised as non-current debt).

Codelco (consolidated)

Continental water (never desalinated)

ENSO-attributable drought years (shaded): 2007, 2010, 2011, 2012, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025



What moved in each ENSO-attributable drought year (year-on-year change)

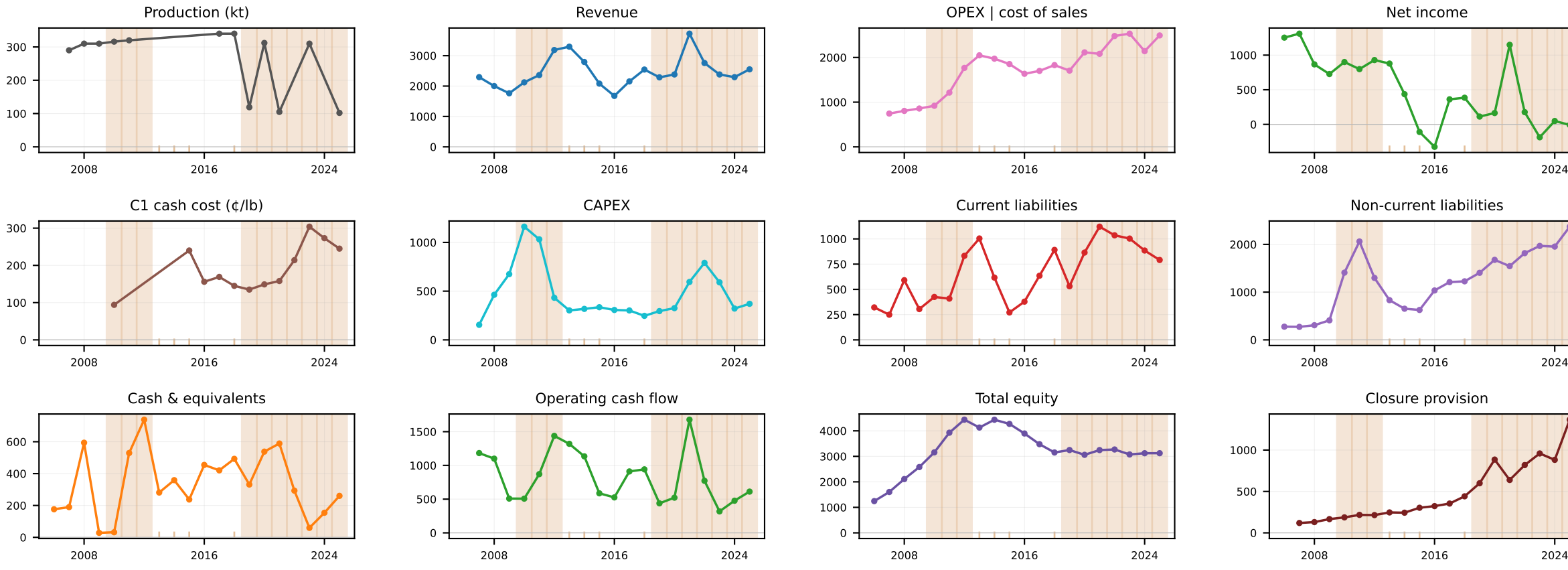
2007 drought. rev -1%, OPEX +10%, NI -11%, CAPEX +32%
2010 drought. rev +30%, OPEX +21%, NI +60%, CAPEX +37%, C1 +12%
2011 drought. rev +9%, OPEX +13%, NI +10%, CAPEX -3%, C1 +11%
2012 drought. prod -0%, rev -9%, OPEX +3%, NI +94%, CAPEX +64%, C1 +40%
2015 drought. prod +5%, rev -15%, OPEX -2%, NI -427%, CAPEX +12%, C1 -8%
2016 drought. prod -4%, rev -1%, OPEX -5%, NI +86%, CAPEX -29%, C1 -9%
2017 drought. prod +9%, rev +27%, OPEX +10%, NI +287%, CAPEX +13%, C1 +8%
2018 drought. prod -0%, rev -2%, OPEX +8%, NI -70%, CAPEX +14%, C1 +2%
2019 drought. prod -7%, rev -12%, OPEX -10%, NI -92%, CAPEX +5%, C1 +2%
2020 drought. prod +1%, rev +13%, OPEX +5%, CAPEX -42%, C1 -9%
2021 drought. prod +1%, rev +48%, OPEX +15%, CAPEX +18%, C1 +3%
2022 drought. prod -17%, rev -19%, OPEX +1%, NI -82%, CAPEX +23%, C1 +25%
2023 drought. prod -8%, rev -4%, OPEX +8%, NI -264%, CAPEX +26%, C1 +23%
2024 drought. prod +4%, rev +4%, OPEX -3%, NI +141%, CAPEX +10%, C1 -2%
2025 drought. rev +15%, OPEX +7%, CAPEX +6%, C1 +5%

ENSO drought year (shaded) | general-drought year | desalination onset

Anglo American Sur

Continental water (never desalinated)

ENSO-attributable drought years (shaded): 2010, 2011, 2012, 2019, 2020, 2021, 2022, 2023, 2024, 2025



What moved in each ENSO-attributable drought year (year-on-year change)

2010 drought. prod +2%, rev +20%, OPEX +7%, NI +24%, CAPEX +72%
2011 drought. prod +1%, rev +11%, OPEX +32%, NI -11%, CAPEX -11%
2012 drought. rev +35%, OPEX +45%, NI +16%, CAPEX -58%
2019 drought. prod -65%, rev -10%, OPEX -7%, NI -71%, CAPEX +20%, C1 -7%
2020 drought. prod +162%, rev +4%, OPEX +24%, NI +45%, CAPEX +10%, C1 +10%
2021 drought. prod -66%, rev +56%, OPEX -2%, CAPEX +83%, C1 +6%
2022 drought. rev -26%, OPEX +19%, NI -85%, CAPEX +33%, C1 +35%
2023 drought. rev -14%, OPEX +2%, NI -205%, CAPEX -25%, C1 +42%
2024 drought. rev -4%, OPEX -15%, NI +126%, CAPEX -46%, C1 -10%
2025 drought. rev +11%, OPEX +16%, NI -117%, CAPEX +15%, C1 -10%

ENSO drought year (shaded) | general-drought year | desalination onset

Los Pelambres

Desalinated from 2024

ENSO-attributable drought years (shaded): 2012, 2014, 2015, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025

What moved in each ENSO-attributable drought year (year-on-year change) (data-error cells removed per audit)

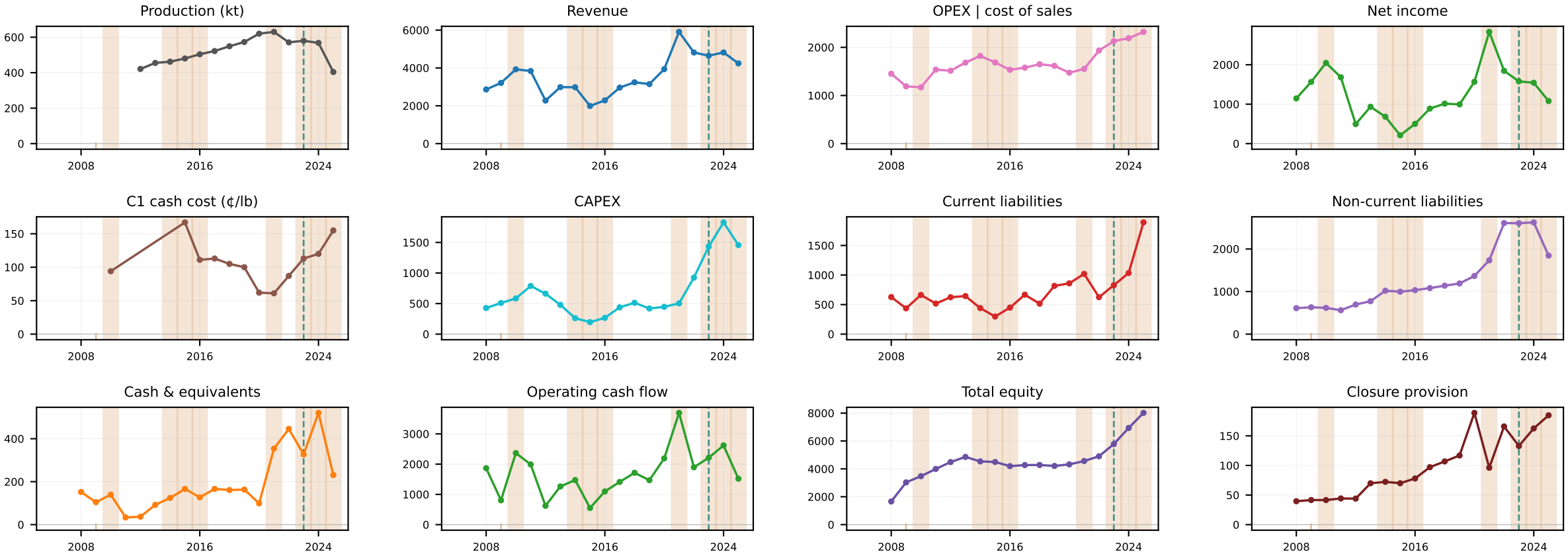
2012 drought. rev -3%, OPEX +7%, NI -11%, CAPEX +24%, C1 +10%
2014 drought. rev -15%, OPEX -13%, NI -26%, CAPEX +10%, C1 +2%
2015 drought. rev -32%, OPEX -16%, NI -57%, CAPEX -18%, C1 +4%
2018 drought. prod +4%, rev +3%, OPEX +9%, NI -8%, CAPEX +51%, C1 -11%
2019 drought. prod +1%, rev -5%, OPEX -4%, NI -2%, CAPEX +48%, C1 +0%
2020 drought. prod -1%, rev +12%, OPEX -21%, NI +20%, CAPEX +54%, C1 -11%
2021 drought. prod -10%, rev +36%, OPEX +11%, NI +64%, CAPEX +8%, C1 +10%
2022 drought. prod -15%, rev -29%, OPEX -7%, NI -47%, CAPEX +7%, C1 +24%
2023 drought. prod +9%, rev +14%, OPEX +14%, NI +17%, CAPEX -5%, C1 +4%
2024 drought. prod +7%, rev +14%, OPEX +38%, NI -12%, CAPEX -7%, C1 +11%
2025 drought. prod -8%, rev +24%, OPEX +9%, NI +39%, CAPEX +29%, C1 -35%

ENSO drought year (shaded) general-drought year desalination onset

Collahuasi

Desalinated from 2023

ENSO-attributable drought years (shaded): 2010, 2014, 2015, 2016, 2021, 2023, 2024, 2025



What moved in each ENSO-attributable drought year (year-on-year change)

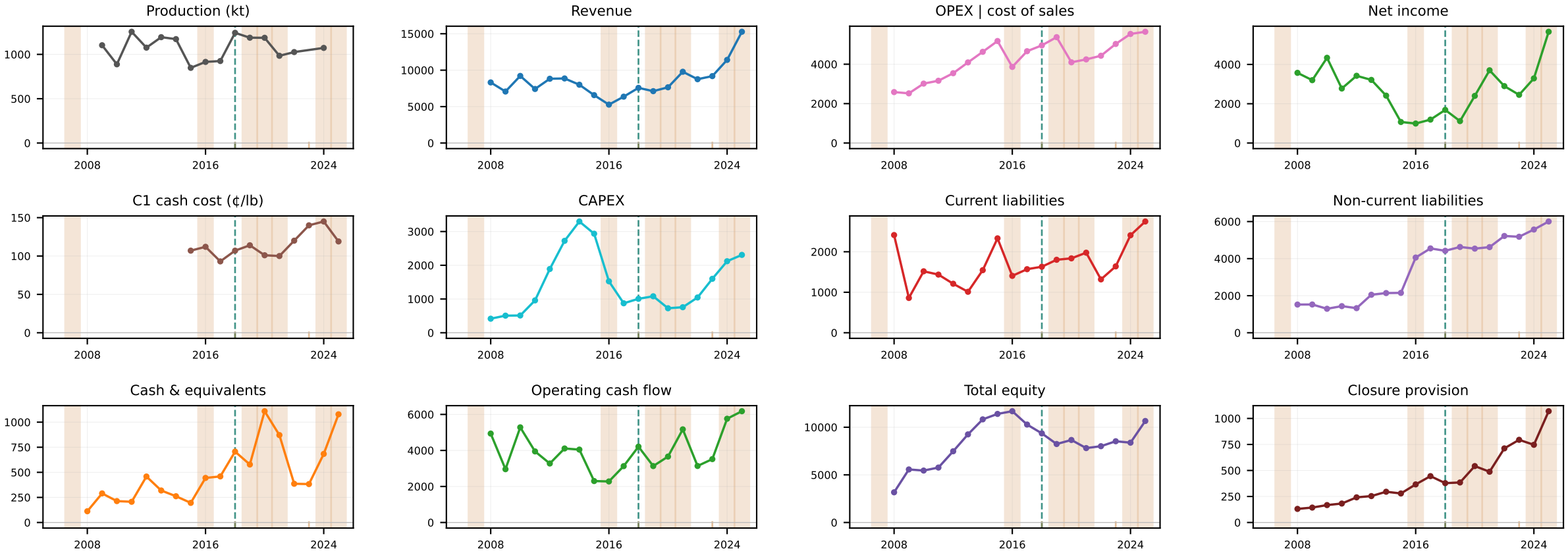
2010 drought. rev +22%, OPEX -2%, NI +31%, CAPEX +15%
2014 drought. prod +2%, rev -0%, OPEX +8%, NI -27%, CAPEX -46%
2015 drought. prod +4%, rev -33%, OPEX -7%, NI -69%, CAPEX -25%
2016 drought. prod +5%, rev +15%, OPEX -9%, NI +137%, CAPEX +36%, C1 -34%
2021 drought. prod +2%, rev +50%, OPEX +5%, NI +81%, CAPEX +13%, C1 -2%
2023 drought. prod +2%, rev -4%, OPEX +10%, NI -14%, CAPEX +55%, C1 +30%
2024 drought. prod -2%, rev +4%, OPEX +3%, NI -2%, CAPEX +28%, C1 +6%
2025 drought. prod -29%, rev -12%, OPEX +6%, NI -30%, CAPEX -20%, C1 +29%

ENSO drought year (shaded) general-drought year desalination onset

Escondida

Desalinated from 2018

ENSO-attributable drought years (shaded): 2007, 2016, 2019, 2020, 2021, 2024, 2025



What moved in each ENSO-attributable drought year (year-on-year change)

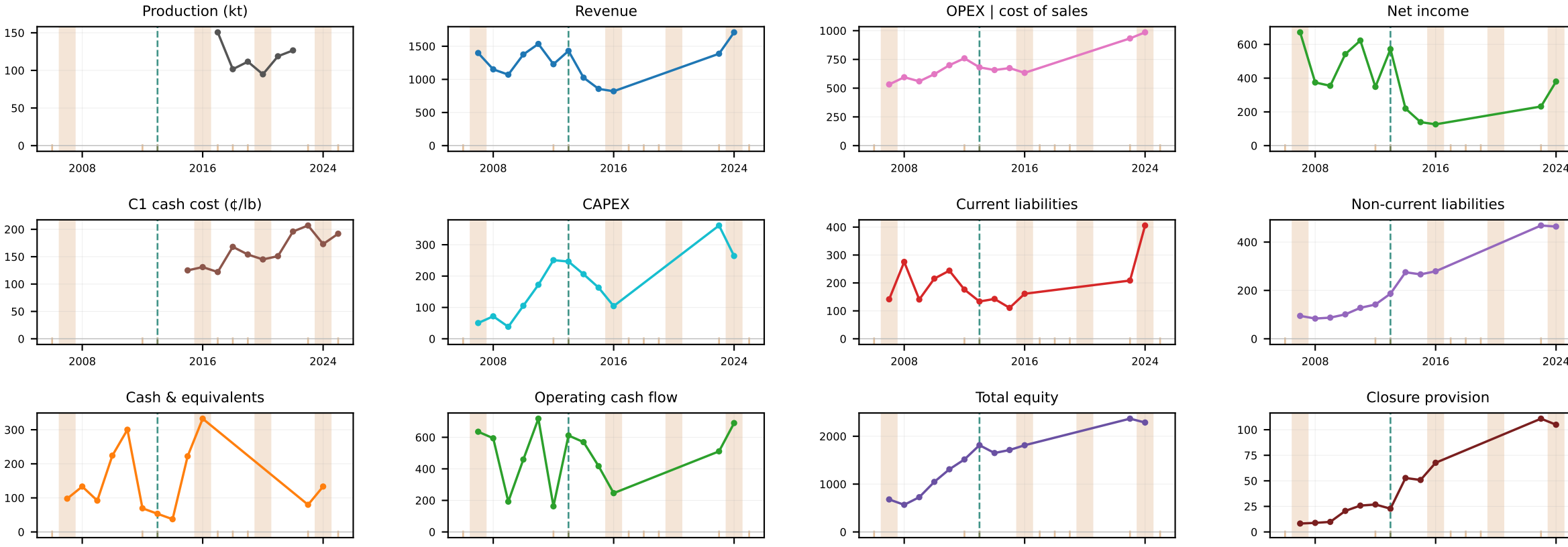
2007 drought. limited disclosure
2016 drought. prod +8%, rev -20%, OPEX -25%, NI -7%, CAPEX -48%, C1 +5%
2019 drought. prod -4%, rev -6%, OPEX +8%, NI -34%, CAPEX +7%, C1 +7%
2020 drought. prod -0%, rev +7%, OPEX -24%, NI +115%, CAPEX -33%, C1 -11%
2021 drought. prod -17%, rev +28%, OPEX +4%, NI +54%, CAPEX +4%, C1 -1%
2024 drought. rev +24%, OPEX +10%, NI +34%, CAPEX +33%, C1 +4%
2025 drought. rev +34%, OPEX +2%, NI +72%, CAPEX +9%, C1 -18%

ENSO drought year (shaded) general-drought year desalination onset

Candelaria

Desalinated from 2013

ENSO-attributable drought years (shaded): 2007, 2016, 2020, 2024



What moved in each ENSO-attributable drought year (year-on-year change)

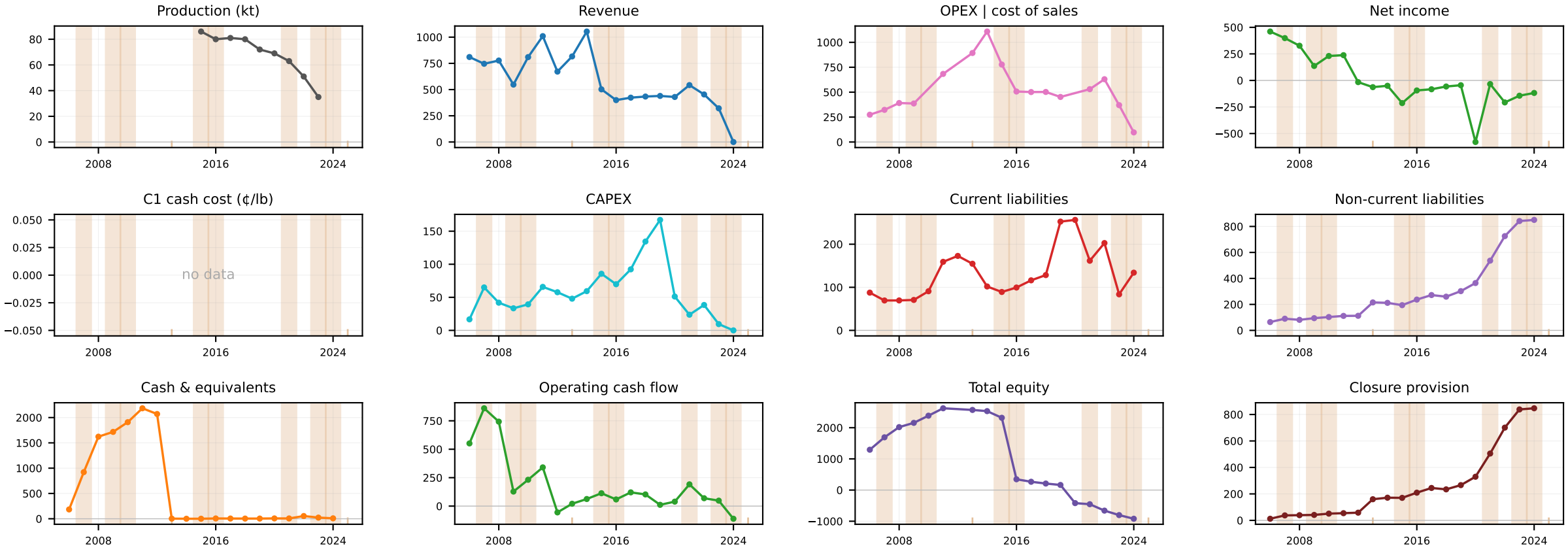
2007 drought. limited disclosure
2016 drought. rev -4%, OPEX -6%, NI -9%, CAPEX -36%, C1 +5%
2020 drought. prod -15%, C1 -6%
2024 drought. rev +23%, OPEX +6%, NI +64%, CAPEX -27%, C1 -16%

ENSO drought year (shaded) general-drought year desalination onset

Cerro Colorado

Continental water (never desalinated)

ENSO-attributable drought years (shaded): 2007, 2009, 2010, 2015, 2016, 2021, 2023, 2024



What moved in each ENSO-attributable drought year (year-on-year change) (data-error cells removed per audit)

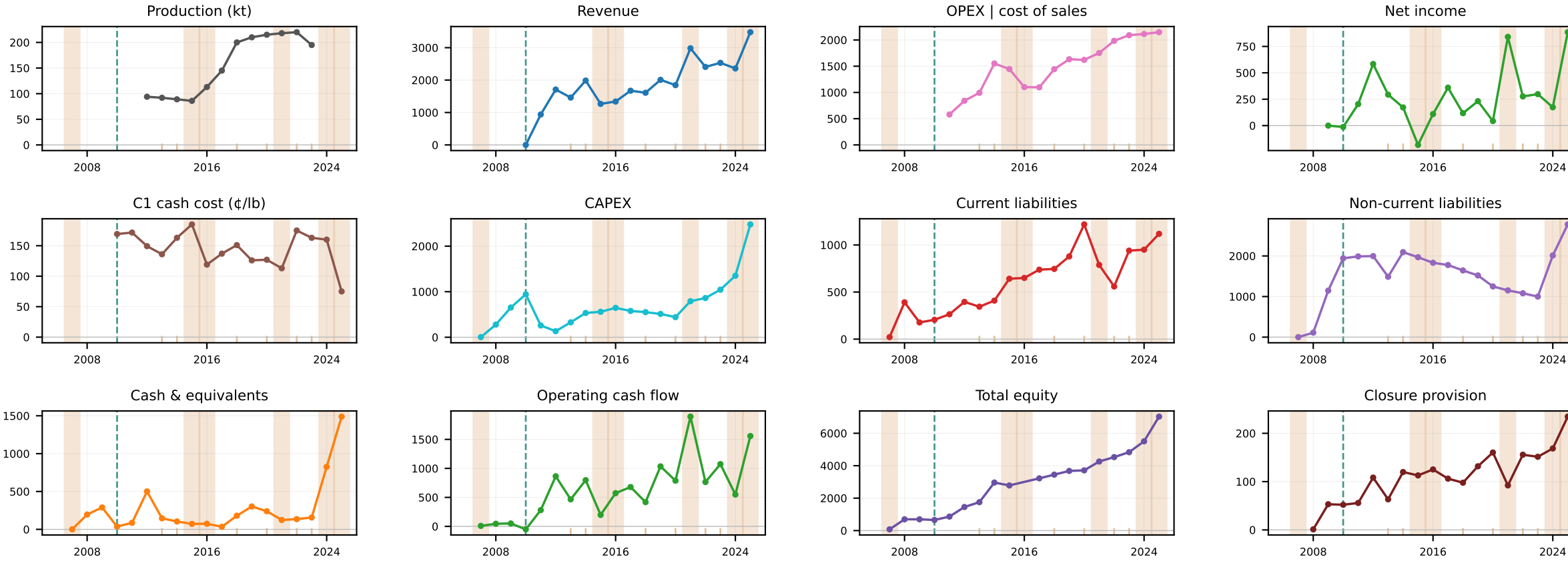
2007 drought. rev -8%, OPEX +18%, NI -13%, CAPEX +289%
2009 drought. rev -30%, OPEX -1%, NI -58%, CAPEX -20%
2010 drought. rev +48%, NI +69%, CAPEX +18%
2015 drought. rev -52%, OPEX -30%, NI -324%, CAPEX +44%
2016 drought. prod -7%, rev -20%, OPEX -35%, NI +56%, CAPEX -18%
2021 drought. prod -9%, rev +26%, NI +94%, CAPEX -54%
2023 drought. prod -31%, rev -29%, OPEX -41%, NI +30%, CAPEX -75%
2024 drought. rev -100%, OPEX -74%, NI +18%, CAPEX -100%

ENSO drought year (shaded) general-drought year desalination onset

Centinela (El Tesoro+Esperanza)

Desalinated from 2010

ENSO-attributable drought years (shaded): 2007, 2015, 2016, 2021, 2024, 2025



What moved in each ENSO-attributable drought year (year-on-year change) (data-error cells removed per audit)

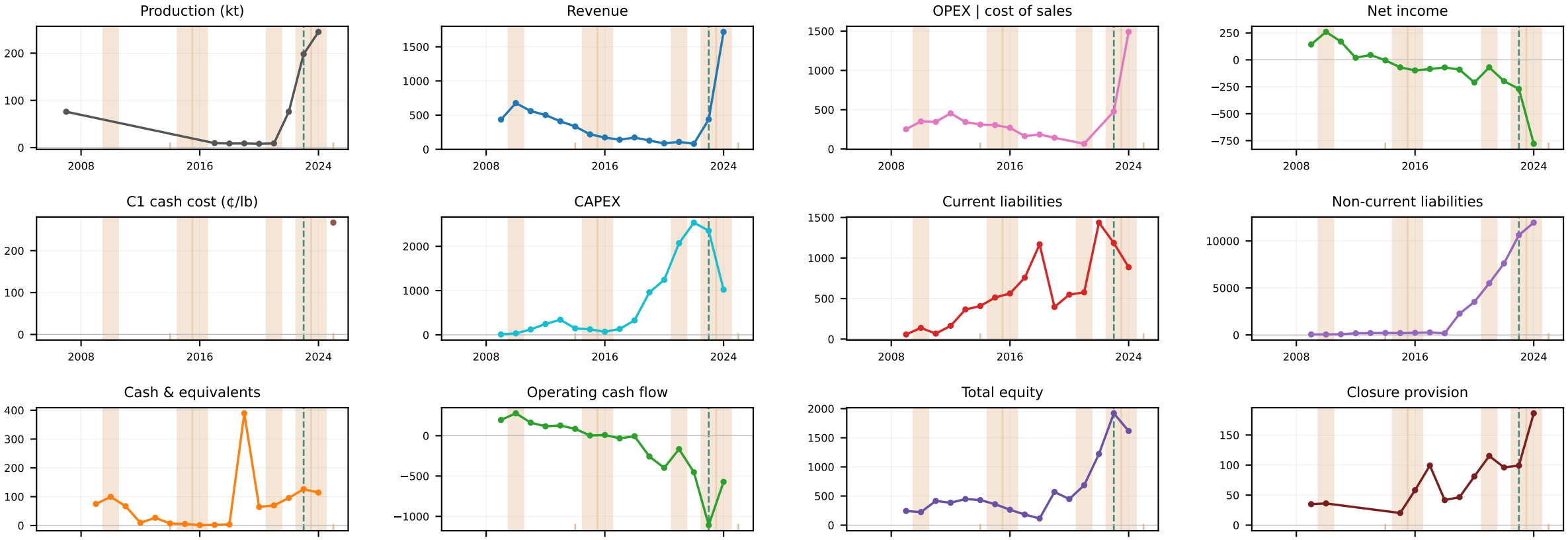
2007 drought. limited disclosure
2015 drought. prod -3%, rev -36%, OPEX -7%, NI -206%, CAPEX +5%, C1 +13%
2016 drought. prod +31%, rev +6%, OPEX -24%, NI +160%, CAPEX +15%, C1 -36%
2021 drought. prod +1%, rev +62%, OPEX +8%, CAPEX +79%, C1 -11%
2024 drought. rev -7%, OPEX +1%, NI -42%, CAPEX +29%, C1 -2%
2025 drought. rev +47%, OPEX +2%, NI +409%, CAPEX +83%, C1 -53%

ENSO drought year (shaded) general-drought year desalination onset

Quebrada Blanca

Desalinated from 2023

ENSO-attributable drought years (shaded): 2010, 2015, 2016, 2021, 2023, 2024



What moved in each ENSO-attributable drought year (year-on-year change) (data-error cells removed per audit)

2010 drought. rev +56%, OPEX +39%, NI +81%, CAPEX +276%
2015 drought. rev -35%, OPEX -2%, CAPEX -15%
2016 drought. rev -21%, OPEX -11%, NI -40%, CAPEX -43%
2021 drought. prod +10%, rev +25%, NI +67%, CAPEX +67%
2023 drought. prod +161%, rev +440%, NI -36%, CAPEX -7%
2024 drought. prod +24%, rev +292%, OPEX +212%, NI -189%, CAPEX -57%

ENSO drought year (shaded) general-drought year desalination onset

El Abra

Continental water (never desalinated)

ENSO-attributable drought years (shaded): 2007, 2010, 2015, 2016, 2021, 2024, 2025

What moved in each ENSO-attributable drought year (year-on-year change)

2007 drought. rev -22%, OPEX -3%, NI -31%
2010 drought. rev +23%, OPEX +60%, NI +3%
2015 drought. prod -3%, rev -29%, OPEX +9%, NI -103%, CAPEX -21%
2016 drought. prod -9%, rev -38%, OPEX -46%, CAPEX -92%
2021 drought. rev +57%, OPEX +8%, CAPEX -27%
2024 drought. rev +7%, OPEX -5%, NI +225%, CAPEX -20%
2025 drought. rev +5%, OPEX +2%, NI +12%, CAPEX +9%

ENSO drought year (shaded) general-drought year desalination onset

Zaldívar

Continental water (never desalinated)

ENSO-attributable drought years (shaded): 2007, 2016, 2020, 2021, 2024

What moved in each ENSO-attributable drought year (year-on-year change)

2007 drought. limited disclosure
2016 drought. prod +275%, rev +229%, OPEX +203%, CAPEX +380%, C1 -10%
2020 drought. prod -17%, rev -13%, OPEX -13%, NI -22%, CAPEX +66%, C1 +3%
2021 drought. prod -9%, rev +42%, OPEX +23%, NI +156%, CAPEX +17%, C1 +33%
2024 drought. prod -1%, rev +0%, OPEX -4%, NI +113%, CAPEX -4%, C1 +2%

ENSO drought year (shaded) general-drought year desalination onset

Spence

Desalinated from 2021

ENSO-attributable drought years (shaded): 2007, 2016, 2020, 2021, 2024

What moved in each ENSO-attributable drought year (year-on-year change) (data-error cells removed per audit)

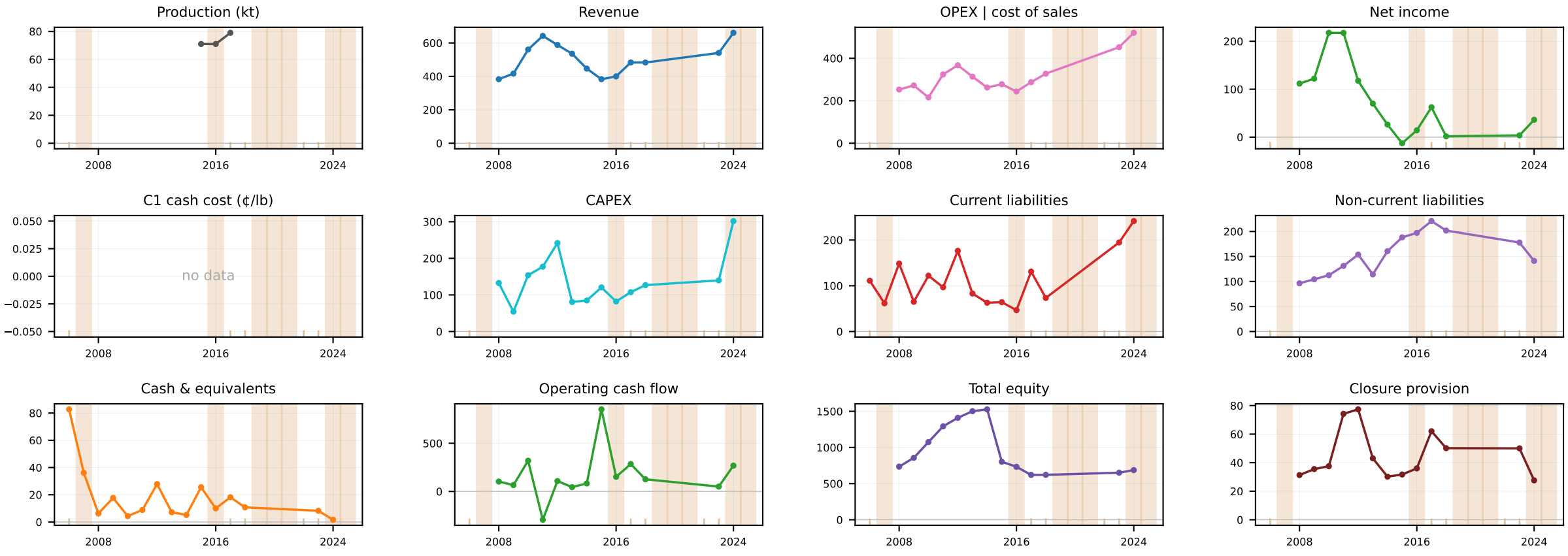
2007 drought. limited disclosure
2016 drought. prod +4%, rev -14%, OPEX +28%, NI -32%, CAPEX +4%
2020 drought. prod -24%, rev -20%, NI -46%, CAPEX -25%
2021 drought. prod +0%, rev +104%, NI +245%, CAPEX -26%
2024 drought. rev +13%, OPEX +10%, NI +37%, CAPEX -8%, C1 +1%

ENSO drought year (shaded) general-drought year desalination onset

Lomas Bayas

Continental water (never desalinated)

ENSO-attributable drought years (shaded): 2007, 2016, 2019, 2020, 2021, 2024, 2025



What moved in each ENSO-attributable drought year (year-on-year change)

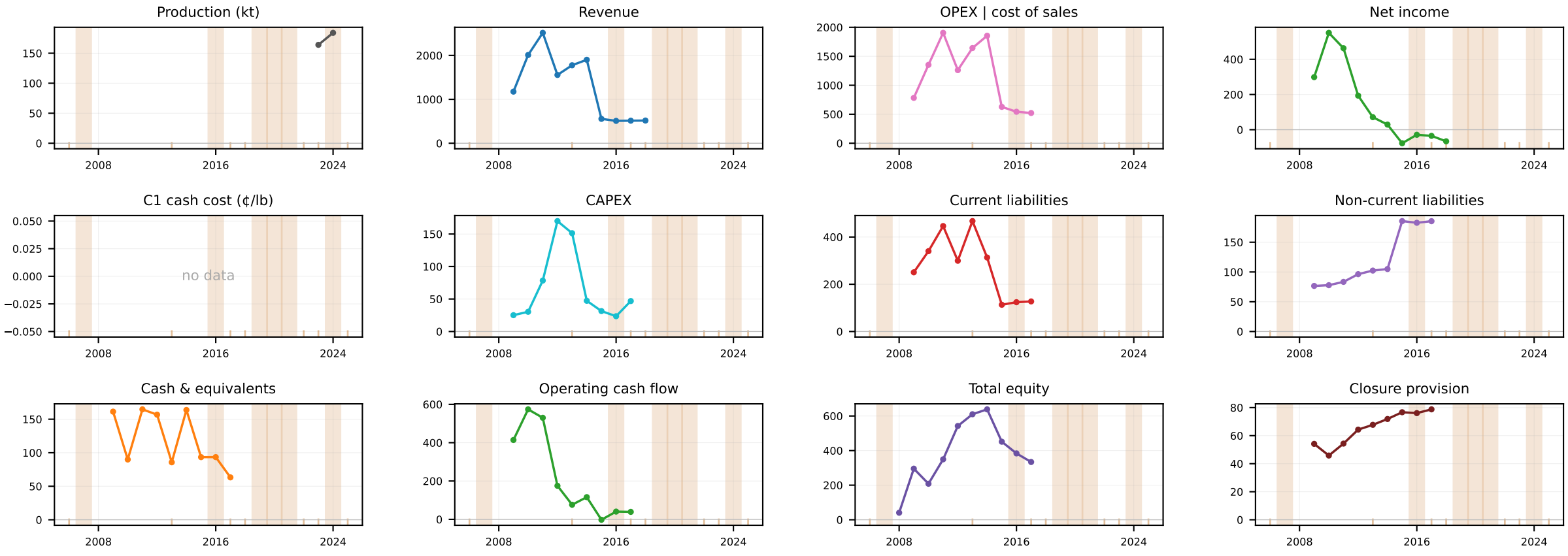
2007 drought. limited disclosure
2016 drought. prod +0%, rev +4%, OPEX -12%, NI +212%, CAPEX -32%
2019 drought. limited disclosure
2020 drought. limited disclosure
2021 drought. limited disclosure
2024 drought. rev +22%, OPEX +15%, CAPEX +116%
2025 drought. limited disclosure

ENSO drought year (shaded) general-drought year desalination onset

Mantos Blancos / Mantoverde

Continental water (never desalinated)

ENSO-attributable drought years (shaded): 2007, 2016, 2019, 2020, 2021, 2024



What moved in each ENSO-attributable drought year (year-on-year change)

2007 drought. limited disclosure
2016 drought. rev -8%, OPEX -13%, NI +63%, CAPEX -25%
2019 drought. limited disclosure
2020 drought. limited disclosure
2021 drought. limited disclosure
2024 drought. prod +12%

ENSO drought year (shaded) general-drought year desalination onset